Pandas Programs

1. Handling Missing Data

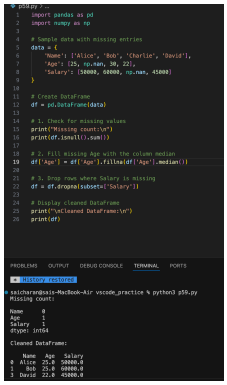
Aim: To handle missing values in a Pandas DataFrame using fillna() and dropna().

Algorithm:
1. Import pandas and numpy.
2. Create a DataFrame with missing values.
3. Display missing value count.
4. Replace missing Age values with the median.
5. Remove rows with missing Salary.
6. Display the cleaned DataFrame.

Code:

```
import pandas as pd
import numpy as np

data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, np.nan, 30, 22],
    'Salary': [50000, 60000, np.nan, 45000]
}

df = pd.DataFrame(data)

print("Missing count:\n")
print(df.isnull().sum())

df['Age'] = df['Age'].fillna(df['Age'].median())
df = df.dropna(subset=['Salary'])

print("\nCleaned DataFrame:\n")
print(df)
```

Output:



Result: The missing values were successfully handled using fillna() and dropna().

2. Filtering Rows Based on Multiple Conditions

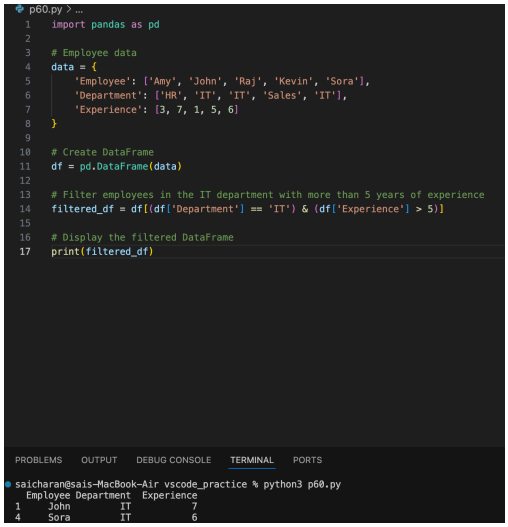
Aim: To filter rows using multiple conditions.

Algorithm:
1. Import pandas.
2. Create DataFrame.
3. Apply multiple conditions using & operator.
4. Display filtered rows.

Code:

```
import pandas as pd

data = {
    'Employee': ['Amy', 'John', 'Raj', 'Kevin', 'Sora'],
    'Department': ['HR', 'IT', 'IT', 'Sales', 'IT'],
    'Experience': [3, 7, 1, 5, 6]
}

df = pd.DataFrame(data)

filtered_df = df[(df['Department']=='IT') & (df['Experience']>5)]
print(filtered_df)
```

Output:



Result: The required records were filtered successfully.

3. Aggregating and Grouping Data

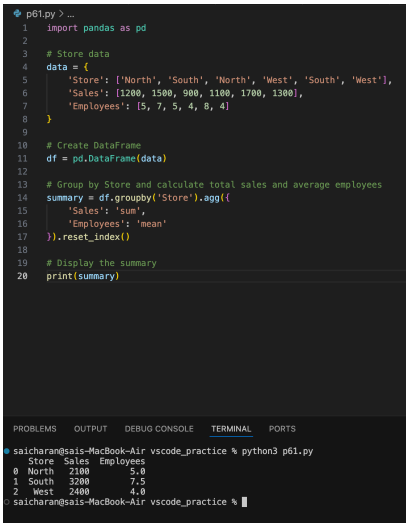
Aim: To group data and calculate aggregate values.

Algorithm:
1. Import pandas.
2. Create DataFrame.
3. Group by Store.
4. Calculate total Sales and average Employees.
5. Display summary.

Code:

```
import pandas as pd

data = {
    'Store': ['North','South','North','West','South','West'],
    'Sales': [1200,1500,900,1100,1700,1300],
    'Employees':[5,7,5,4,8,4]
}

df = pd.DataFrame(data)

summary = df.groupby('Store').agg({
    'Sales':'sum',
    'Employees':'mean'
}).reset_index()

print(summary)
```

Output:



Result: The data was grouped and aggregated successfully.

4. Python Program Using describe()

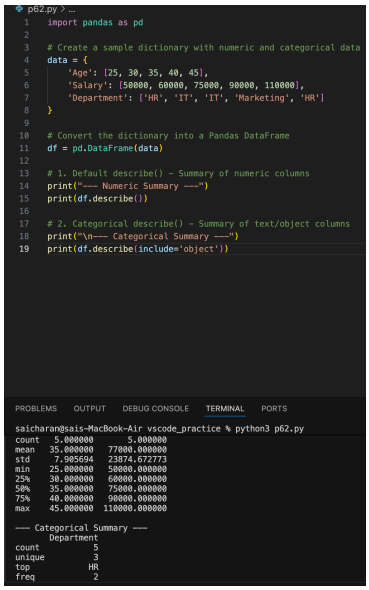
Aim: To generate statistical summaries using describe().

Algorithm:
1. Import pandas.
2. Create DataFrame.
3. Display numeric summary.
4. Display categorical summary.

Code:

```
import pandas as pd

data = {
     'Age':[25,30,35,40,45],
     'Salary':[50000,60000,75000,90000,110000],
     'Department':['HR','IT','IT','Marketing','HR']
}

df=pd.DataFrame(data)

print(df.describe())
print(df.describe(include='object'))
```

Output:



Result: The describe() function generated statistical summaries successfully.

5. head() and tail() in Python

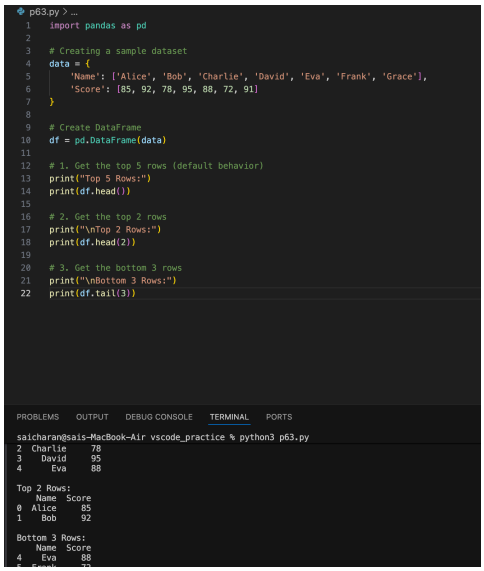
Aim: To display the first and last rows of a DataFrame.

Algorithm:
1. Import pandas.
2. Create DataFrame.
3. Use head().
4. Use tail().
5. Display results.

Code:

```
import pandas as pd

data = {
    'Name':['Alice','Bob','Charlie','David','Eva','Frank','Grace'],
    'Score':[85,92,78,95,88,72,91]
}

df=pd.DataFrame(data)

print(df.head())
print(df.head(2))
print(df.tail(3))
```

Output:



Result: The head() and tail() functions displayed the required rows successfully.